paper_id: PAPER_926
title: "Multi-AGN Monte Carlo Jet Power Batch"
session: 211
date: 2026-04-11
author: "Daniel T. Murphy"
status: production
cvw: "v2.0.0"
tags: [quasar, galaxy, AGN, SCm, jet, phonon, UQFF]
sm_anchor: "CVW v2.0.0 — G6 SM Anchor Gate compliant"

PAPER_926: Multi-AGN Monte Carlo Jet Power Batch

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-11 **Session:** 211 **Source:** SCm phonon gap implementation (agn_{jet_power_curves}.py) **Calculator:** MultiAGNJetPowerMonteCarloBatchCalc (CP4 #510) **CVW:** v2.0.0 compliant

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Abstract

Monte Carlo sampling (10^6 samples) of phonon-enhanced jet power P_jet across four AGN archetypes: 3C273-type (M = 8.86e8 M_sun), CenA-type (5.5e7 M_sun), TON618-type (6.6e10 M_sun), and TXS0506-type (3e8 M_sun). Gamma draws from N(mu_Gamma, sigma_Gamma^2) with mu = 1.25 THz, sigma = 0.15 THz to model thermal phonon linewidth fluctuations. Reports mean, standard deviation, median, p5, and p95 percentiles of P_jet(Gamma) for each system. The P_BZ baseline scales as M^2, giving a dynamic range of ~10^8 in base jet power across the four systems.

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1. Core Equations

Section A: Lagrangian

$$P_jet = P_BZ * (1 + M_jet(Gamma))$$
$$M_jet(Gamma) = 1 + A_jet * exp[-(Gamma - Gamma_0)^2/(2 * sigma_Gamma^2)]$$
$$P_BZ = (pi/(6 * mu_0)) * B^2 * r_g^2 * c * a^2$$
$$Gamma\ N(1.25THz, 0.15^2THz^2)$$

Section B: VDS/DVP/BH Number Systems

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MC statistics: mean(P_jet), std(P_jet), median(P_jet), P_5, P_95
Modulation range: [min(1+M_jet), max(1+M_jet)] over all draws
System diversity: P_BZ(TON618) / P_BZ(CenA) ~ (6.6e10/5.5e7)^2 ~ 1.4e6
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Section SM: SM Anchors

$$3C273 - type : M = 8.86e8 M_sun, high - luminosity quasar$$
$$CenA - type : M = 5.5e7 M_sun, nearby radio galaxy$$
$$TON618 - type : M = 6.6e10 M_sun, ultramassive quasar$$
$$TXS0506 - type : M = 3e8 M_sun, blazar neutrino source$$

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2. Parameters

Parameter	Default	Description
`M_{bh\_Msun}`	8.86e8	BH mass (solar masses)
a_spin	0.9	Spin parameter
B_field	50 T	Magnetic field
A_jet	1.5	Modulation amplitude
`sigma_{Gamma\_THz}`	0.08	sigma_Gamma
`Gamma_{mean\_THz}`	1.25	MC mean Gamma
`Gamma_{std\_THz}`	0.15	MC std dev
n_samples	100000	MC sample count

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3. Key Results

System	P_BZ (erg/s)	(erg/s)	Spread
3C273-type	~10^46	~2.5*P_BZ	+/-30%
CenA-type	~10^42	~2.5*P_BZ	+/-30%
TON618-type	~10^50	~2.5*P_BZ	+/-30%
TXS0506-type	~10^44	~2.5*P_BZ	+/-30%

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4. Physical Interpretation

The MC approach captures the stochastic nature of phonon linewidth fluctuations in real AGN environments. The 30% spread in P_jet at fixed BH mass and spin arises entirely from phonon frequency variations, providing a natural explanation for AGN jet power variability on timescales of hours to days (matching observed VHE flaring). The consistent ~2.5x mean enhancement across all four systems demonstrates that phonon modulation is mass-independent, affecting only the multiplicative factor while P_BZ provides the mass-dependent baseline.

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5. References

- PAPER_920: Monte Carlo jet power sampling (Session 210c)
- PAPER_925: Quasar jet phonon modulation M_jet(Gamma)
- agn_{jet_power_curves}.py: 3-class standalone module

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_{U_Bi_i} jet modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\mathrm{Edd}}^{\mathrm{UQFF}} = L_{\mathrm{Edd}} \cdot \left(1 + \frac{\rho_{\mathrm{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{GM/r_H^2}\right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c/\sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37}$ J/m^3 is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> *Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \tfrac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37}$ J/m^3 (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda}\, v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 5970$ GeV (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_{U_Bi_i} buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4}\,\text{day}^{-1}$	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system

Constant	Symbol	Value	Validation Domain
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25$ THz	Phonon resonance
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37}\,\mathrm{J/m^3}$	Fundamental

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Phonon frequency ω_{SCm}	$2\pi \times 1.25$ THz (Pd-D lattice)	Measured Pd-D phonon spectrum	Fukai (2005)	Mapped to SCm
Vacuum energy ρ_{vac}	7.09×10^{-37} kg/m^3	$\rho_{\mathrm{vac}} \sim 10^{-29}$ g/cm^3	Planck 2018	Novel SCm scale
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%

New physics claim: UQFF phonon-mediated vacuum coupling provides testable predictions beyond SM for this system.

Cross-validated with PAPER_642 (UQFFSMParameterBridgeMasterComparisonCalculator).

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§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: SCm-phonon (lattice resonance)

§A.2 Lagrangian Density

$$\mathcal{L}_{SCm\_phonon} = \sum_{i=1}^{26} [U_{g,i} + U_{m,i} + U_{A,i} - U_{b,i}] \cdot S_{26}([SSq]) \cdot \Phi_{1.25\mathrm{THz}}(\omega, \Gamma)$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_m u \frac{\partial \mathcal{L}}{\partial(\partial_m u \phi)} = 0 \implies F_{U,Bi\_i} = -\nabla U_{\mathrm{eff}} + \Phi \cdot S_{26} \cdot E_{\mathrm{net}}}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms → SCm vacuum → phonon ω_{SCm} → lattice resonance → F_{U,Bi_i} unified force → observational prediction

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Appendix: Session 225 Cross-References (PAPER_1000–1081)

> *Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).*

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_{U_Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U_Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified

15 cross-reference(s) identified.

References

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Schmidt, M. (1963). *3C 273: A star-like object with large red-shift.* Nature **197**, 1040 — doi:10.1038/1971040a0
4. Richards, G.T. et al. (2006). *The Sloan Digital Sky Survey Quasar Survey.* AJS **166**, 470 — arXiv:astro-ph/0601434 — doi:10.1086/506525
5. de Vaucouleurs, G. (1948). *Recherches sur les Nebuleuses Extragalactiques.* Ann. Astrophys. **11**, 247
6. Kennicutt, R.C. & Evans, N.J. (2012). *Star Formation in the Milky Way and Nearby Galaxies.* ARA&A **50**, 531 — arXiv:1204.3552 — doi:10.1146/annurev-astro-081811-125610
7. Sofue, Y. & Rubin, V. (2001). *Rotation Curves of Spiral Galaxies.* ARA&A **39**, 137 — arXiv:astro-ph/0010594 — doi:10.1146/annurev.astro.39.1.137
8. Fabian, A.C. (2012). *Observational Evidence of Active Galactic Nuclei Feedback.* ARA&A **50**, 455 — arXiv:1204.4114 — doi:10.1146/annurev-astro-081811-125521
9. McNamara, B.R. & Nulsen, P.E.J. (2007). *Heating Hot Atmospheres with Active Galactic Nuclei.* ARA&A **45**, 117 — arXiv:0709.4098 — doi:10.1146/annurev.astro.45.051806.110625
10. Heckman, T.M. & Best, P.N. (2014). *The Coevolution of Galaxies and Supermassive Black Holes.* ARA&A **52**, 589 — arXiv:1403.4620 — doi:10.1146/annurev-astro-081913-035722
11. Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem.* Stud. Hist. Phil. Mod. Phys. **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3
12. Weinberg, S. (1989). *The Cosmological Constant Problem.* Rev. Mod. Phys. **61**, 1 — doi:10.1103/RevModPhys.61.1
13. Blandford, R.D. & Znajek, R.L. (1977). *Electromagnetic extraction of energy from Kerr black holes.* MNRAS **179**, 433 — doi:10.1093/mnras/179.3.433
14. Blandford, R.D. & Payne, D.G. (1982). *Hydromagnetic flows from accretion discs and the production of radio jets.* MNRAS **199**, 883 — doi:10.1093/mnras/199.4.883

15. Ashcroft, N.W. & Mermin, N.D. (1976). *Solid State Physics.* Harcourt
16. Kittel, C. (2004). *Introduction to Solid State Physics.* 8th ed. Wiley
17. Feynman, R.P. (1982). *Simulating Physics with Computers.* Int. J. Theor. Phys. **21**, 467 — doi:10.1007/BF02650179